



Crochet Design Collection

By,

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RAT-IT-2324-F-0080

**A report submitted in partial fulfillment of the requirements for
the higher national diploma in Information Technology**

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Sri Lanka Institute of Advanced Technological Education

2024

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THE REPORT

Submitted in partial fulfillment of the requirements

Of

HIGHER NATIONAL DIPLOMA IN INFORMATION TECHNOLOGY

2024

Advanced Technology Institute,

Rathnapura,

Sri Lanka

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Declaration

I declare that this report is our own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education. Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

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.....

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Dedication

I dedicate this endeavor to my mentors, family, and friends, whose help and advice have been crucial to my success.

I hope this initiative opens up new opportunities in the crochet and technology worlds for everyone who values creativity and innovation.

Acknowledgements

I want to sincerely thank my supervisors, Mr. D.L.C.I. Atukorala, Mr. Yasitha Dissanayaka, and Mr. G.S.T. Bandara, for their tremendous assistance, encouragement, and direction during this project. Their knowledge and helpful criticism have greatly influenced the creation of this Crochet Design Collection Website & Database System.

I also want to express my sincere gratitude to the company's owner, Mrs. A.K.N.M. Gunathilaka, for her collaboration and insightful advice, which greatly aided this project's practical elements.

Lastly, I want to express my gratitude to my family, friends, and coworkers for your constant encouragement and support throughout this journey.

Thank You,

A.B.M.C.U.Basnayaka

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Abstract

Currently run through a Facebook page, the Crochet Design Collection Website is an online platform created to increase the legitimacy and visibility of a crochet business. The website tackles important issues like low consumer trust, limited search exposure, and Facebook's policy limitations. This project makes use of Draw.io for sophisticated database system design, MS Access for database administration, and Ecwid as an e-commerce platform to effectively develop the website. For UI/UX design, branding, and documentation, other technologies including Figma, Canva, and Office Package were utilized. The development process, tools utilized, difficulties encountered, testing techniques, and potential future enhancements are described in this paper.

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Chapter 1: Introduction

1.1 Project Overview:

Crochet Design Collection Website provides an efficient and effective way for you to go about selling your crochet products. At the beginning, the business was marketed and sold using a Facebook page, but it was not visible, trustworthy, and scalable enough. As part of this project, I will create my own store e-commerce website with Ecwid, connect it with a powerful MS Access database for handling products, and develop all interface contents in Figma and Canva.

1.2 Objectives

- · Build an easy to use, scalable e-commerce website with Ecwid.
- · Create a structured and powerful database system in MS Access.
- · Enhance site designs and usability with Figama and Canva
- You can use the intuitive Admin panel for product management and customer interaction.
- Custom database solution for better systemization of data and tracking of handling.
- The main objective is to provide business owner with shopping cart, a easy to use online store to sell their product and service.
- Shows the information and description of the Woolean cloth
- Editing, adding and updating of all records
- Manage the details
- Provide fast accurate services
- Provide services to customer
- Provide services to administrator
- Provide the searching facilities

1.2.1 Crochet Design Collection's Mission

Crochet Design Collection's aim is to design the most practical, most wearable item in your wardrobe

1.2.2 Crochet Design Collection's Vision

Wool is the foundation of our brand; it equips us to do the things we love

Chapter 2: Tools and Technologies Used

2.1 Software and Tools:

2.1.1 Software

2.1.1.1 Ecwid

- **Purpose:** E-commerce platform to manage your online store.
- **Contribution to the Project:** Ecwid lets you create a working online store where customers may peruse, pick, and buy crochet patterns. It is an essential component of the capability for selling your products since it facilitates connectivity with your website, controls inventory, and processes payments.

2.1.1.2. MS Access

- **Purpose:** Database management software.
- The project's databases are created and managed using Microsoft Access. It facilitates the organization of data, including orders, transactions, product details, and customer information. Your website will be efficient and well-structured if you use Microsoft Access to store, query, and report this data.

2.1.1.3. Office Package (Microsoft Word, Excel, PowerPoint, etc.)

- **Purpose:** General office productivity suite.
- You can utilize the Office suite to create written reports, presentations, and documentation for your project. Excel can be used for data analysis, Microsoft Word will be utilized to compose your final report, and PowerPoint may be useful for project presentations or showing your collection of designs.

2.1.1.4. Figma

- **Purpose:** Collaborative design tool for UI/UX.
- Your website's interface and layout are designed using Figma. It enables you to produce wireframes, mockups, and prototypes that specify the site's interactive and visual components. It is very useful for organizing the user experience (UX) and making sure the website functions and looks well for users.

2.1.1.5. Canva

- **Purpose:** Graphic design tool for creating marketing materials.
- Canva assists you in creating eye-catching marketing collateral including posters, banners, and social media postings. By showcasing your crochet design collection to prospective buyers, these images will aid in the branding and marketing initiatives of your website.

2.1.1.6. Draw.io

- **Purpose:** Diagramming and flowchart tool.
- Draw.io is used for producing visual representations of processes, such as website flowcharts, user journeys, and system diagrams. Clear communication with your team and stakeholders will be facilitated by these diagrams, which will help you visualize the organization of your website and the data flow within the system.

2.1.2 Hardware Tools

- Processor – Intel(R) Core(TM) i5-2500 CPU @ 3.30GHz 3.30 GHz
- System type - 64-bit operating system, x64-based processor
- RAM - 2.00 GB

2.2 Justification of Tools

- **Ecwid:** Facilitates e-commerce connectivity for order management and crochet design sales.
- **MS Access:** Oversees databases used to store inventory, orders, and customer data.
- **Office Package:** Throughout the project, this package is used for reports, presentations, and documentation.
- **Figma:** Creates wireframes, prototypes, and website layouts to guarantee a seamless user experience.
- **Canva:** Produces graphic marketing collateral to advertise the crochet patterns.
- **Draw.io:** Aids in the creation of data flow and website structure diagrams for improved project comprehension.

Chapter 3: Project Design and Development Process

3.1 Planning:

Prior to beginning any action, planning is crucial. And success is assured if it is well-planned. Three main administration components make up the community forum. User interaction and moderation. We made plans based on my analysis of the overall difficulty of each of these components, which indicated that the project would take about 16 weeks to finish.

When organizing the many stages of our project, we made the decision to adhere to the Software Development Life Cycle, or SDLC. The following tasks make up this approach.

1. Initial inquiry
2. Establishing the needs for the system
3. Analysis of systems
4. System design
5. Software development
6. System testing
7. Execution and assessment

In the first step, we conducted an analysis of the system to determine its market requirements, its market value, and how it will benefit the user. I've been learning JSP fundamentals for the first three weeks. Primary investigation is ongoing in the interim. We then allowed about two weeks for system design in order to gather requirements and conduct system analysis.

Deliverable and milestone Management must be informed. This information can only be provided as a document that describes the current status of the software being produced because software is intangible. A project's planning involves setting a number of milestones.

3.1.1 Milestone

- Milestone is an end-point of the software process activity.
- At each Milestone there should be formal output that can be represented to the management.
- Milestone output need not be large document; they are the short output of achievements in software project activity.

3.1.2 Deliverable

- Deliverable is a project report that is delivered to the customer.
- Deliverables are delivered to the customer at the end of some major project phase such as specification, design, etc.

3.1.3 Time Line:

Project Plan

Time Line



Name	Start	End	Duration	August	September	October	November
Project Proposal	31.08.2024	04.09.2024	01 Week				
Planning	05.09.2024	11.09.2024	01 Week				
Design	11.09.2024	23.09.2024	02Weeks				
Development	24.09.2024	11.10.2024	02Weeks				
Testing	12.10.2024	24.10.2024	02Weeks				
Error Handling	25.10.2024	31.10.2024	01 Week				
Launch	01.11.2024	07.11.2024	01 Week				

Table-01
Time Line

3.1.4 ER Diagram :

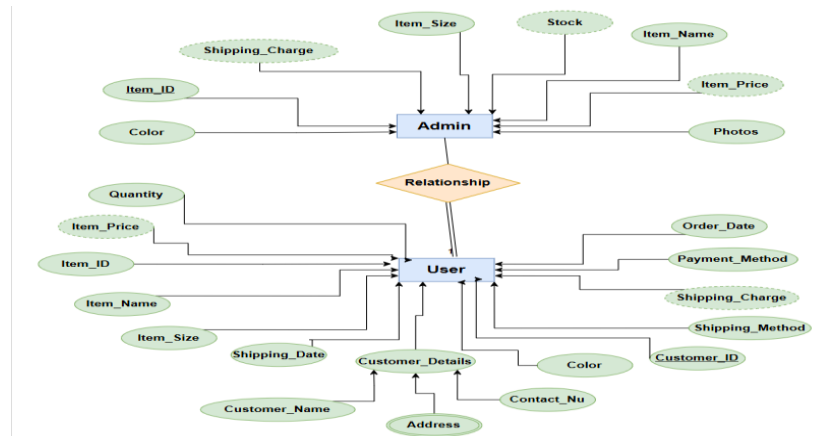


Figure 01- ER Diagram

3.1.5 Wire Frame :

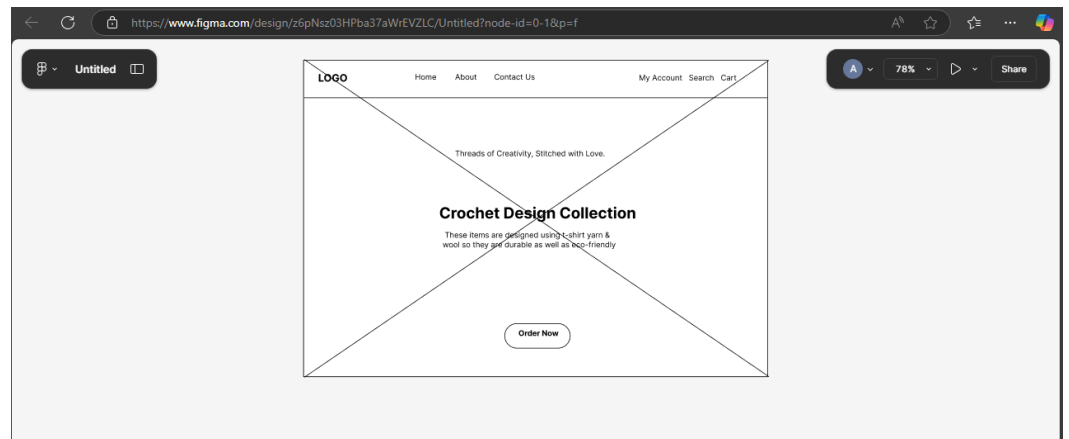


Figure 02- Wire Frame

3.1.6 Budget Plan

3.1.6.1 Technical Cost

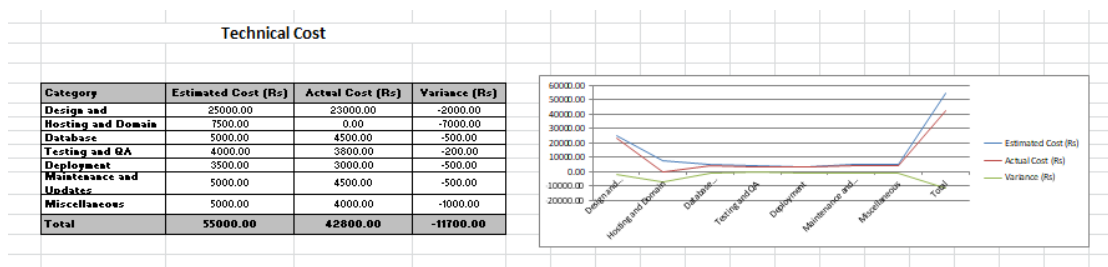


Table-02 Technical Cost

3.1.6.2 Monthly Cost

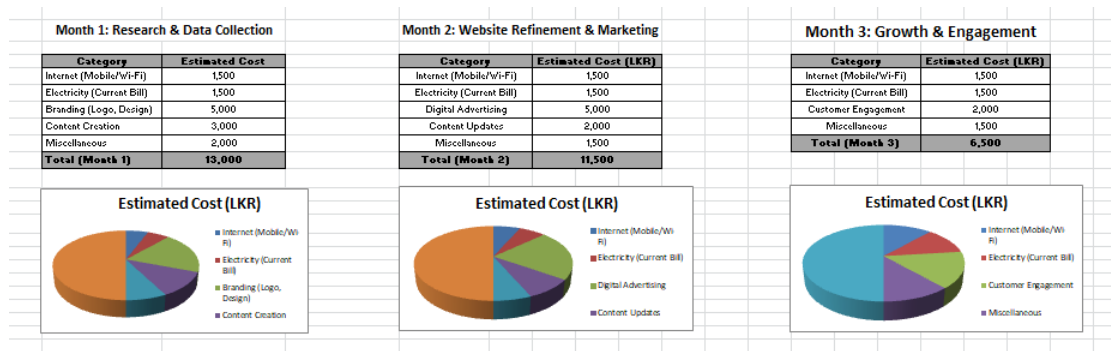


Table-02 Monthly Cost

3.1.6.3 User Case

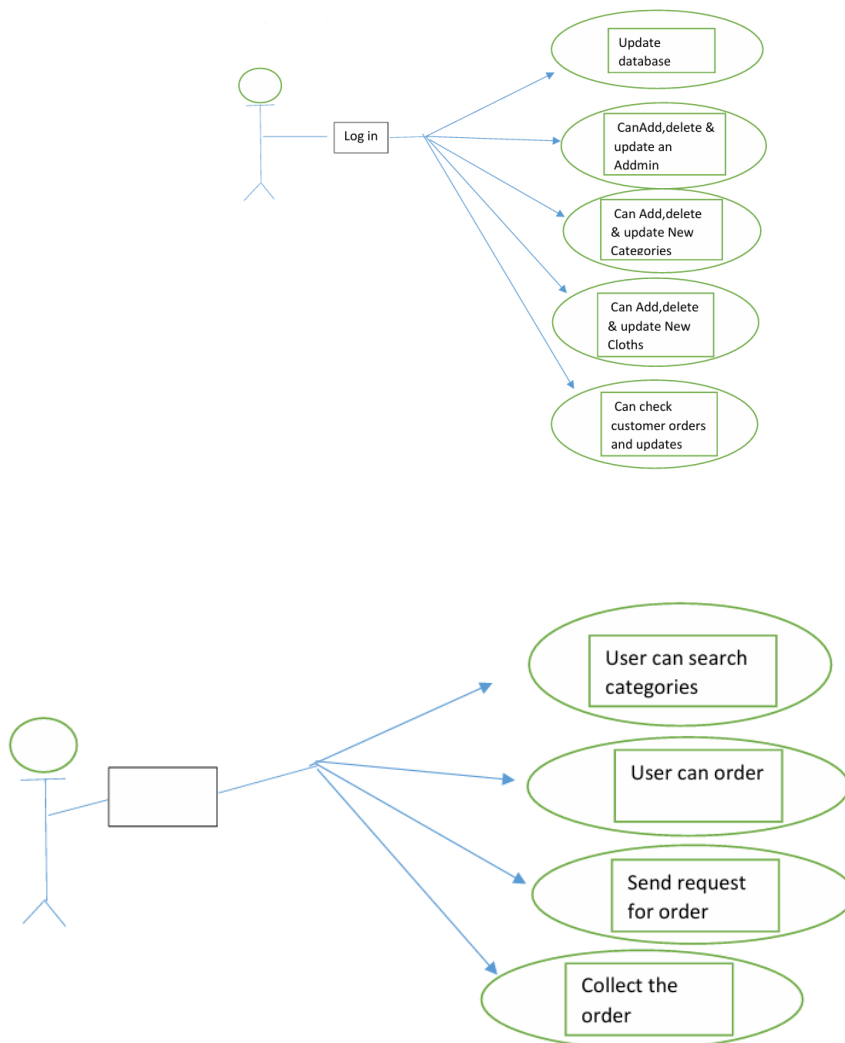


Figure 03- User Case - Admin

Figure 04- User Case - User



Chapter 4: Key Features and Functions

4.1 Feature Summary

A number of cutting-edge elements are included into the Crochet Design Collection website to guarantee a seamless and intuitive buying experience. Among these features are:

- **E-commerce Platform Integration:** Ecwid was used in the development of the website, enabling secure checkout, easy-to-use shopping cart functionality, and seamless product administration.
- **Detailed Product Catalog:** Customers can peruse an orderly assortment of crochet patterns arranged according to price, popularity, and type. High-quality photos, descriptions, prices, and availability information are all included on each product page.
- **Advanced Database administration:** To ensure smooth operations and improved data administration, product details, customer information, and order history are effectively stored and retrieved using an organized MS Access database.
- **User Registration and Profiles:** Clients can register in order to manage their personal information, track their orders, and store their favorite products.
- **Secure Payment Processing:** To ensure secure transactions, the system accepts a number of payment methods, such as PayPal, credit/debit cards, and other online payment gateways.
- **Order tracking and notifications:** Clients are notified in real time via email or SMS regarding the status of their orders, shipment information, and anticipated delivery dates.
- **Search and Filtering Features:** Customers may quickly look for products by utilizing keywords or by category, availability, or price range.
- **Customer Reviews and Ratings:** By allowing consumers to rate and evaluate products, future buyers will be better able to make selections.

- **FAQs and Live Chat Support:** Customers can get real-time assistance via a dedicated chat support function. Answers to frequently asked questions are also available in a thorough FAQ section.
- **User-Friendly Admin Panel:** Store owners can track customer orders, manage inventory, change product data, and keep an eye on sales with this admin panel.
- **Mobile-Responsive Design:** The website offers a flawless shopping experience on all platforms because it is entirely suited for users of smartphones and tablets.
- **Marketing and SEO Tools:** Integrated SEO tools raise the website's search engine rating. Features like email marketing and social media aid in drawing in and keeping clients.
- **Customizable Themes and Layouts:** The website's interface is made to be aesthetically pleasing and adaptable to brand identity using Figma and Canva.
- **Automated Discount and Coupon System:** To improve consumer engagement and boost sales, store owners can generate coupon codes, seasonal promotions, and promotional discounts.
- **Inventory Management System:** By providing customers with up-to-date information on product availability, real-time stock updates help to minimize order cancellations brought on by out-of-stock items.

Together, these characteristics produce a very useful, safe, and easy-to-use online store that increases client satisfaction and operational effectiveness.

4.2 Screenshots and Illustrations

4.2.1 Web Site



Figure 06- Home Page

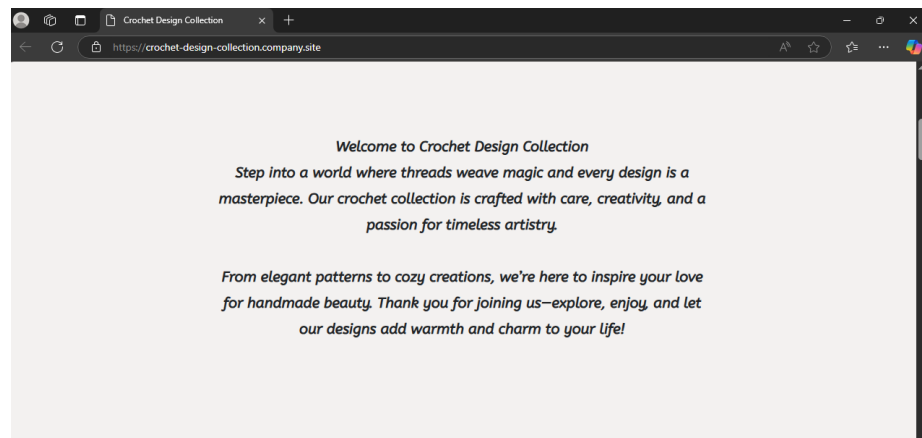


Figure 07- About Page

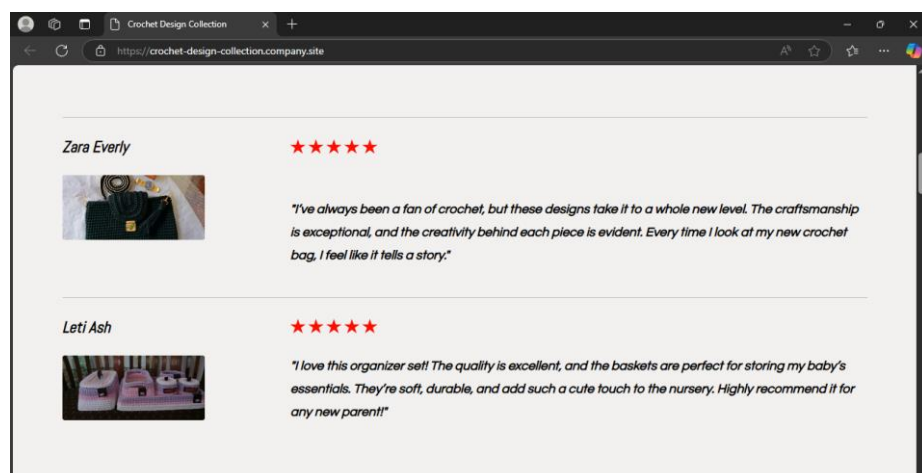


Figure 08- Feed Back Page

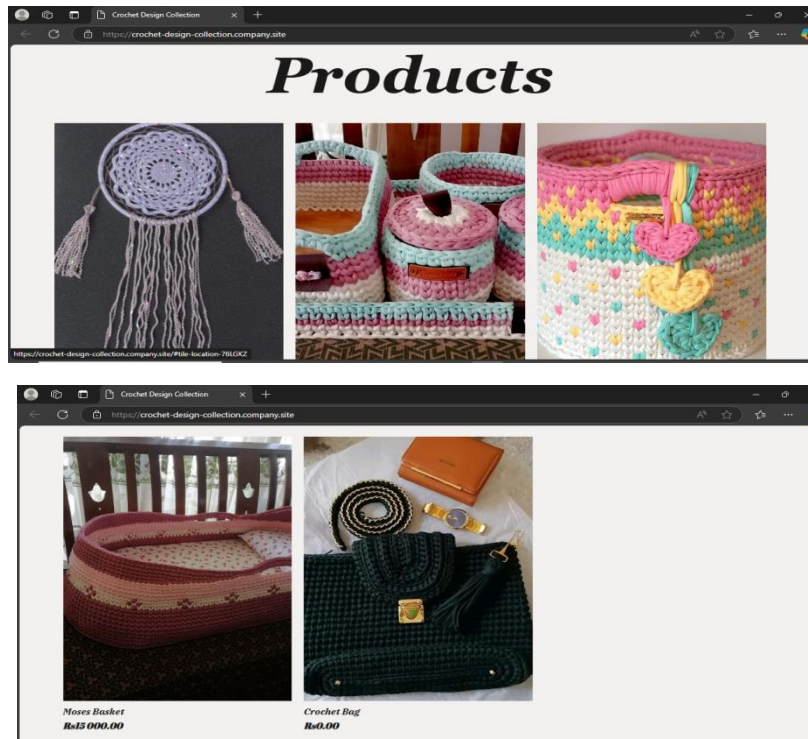


Figure 09- Product

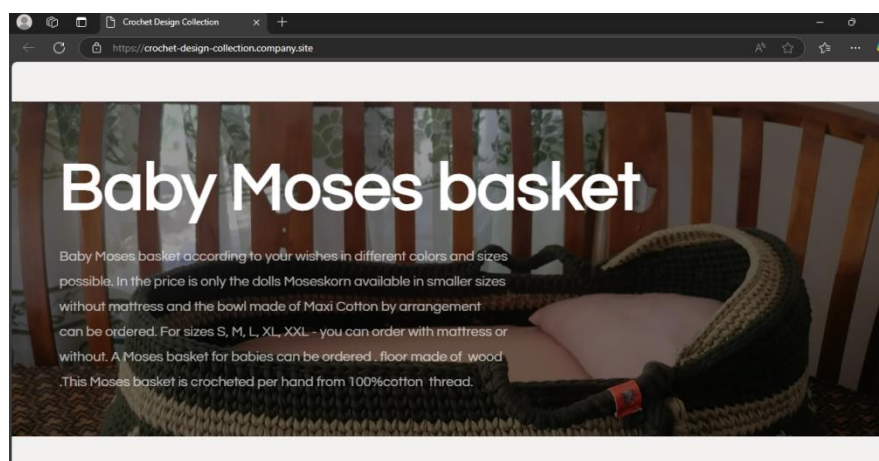


Figure 10- Best Product

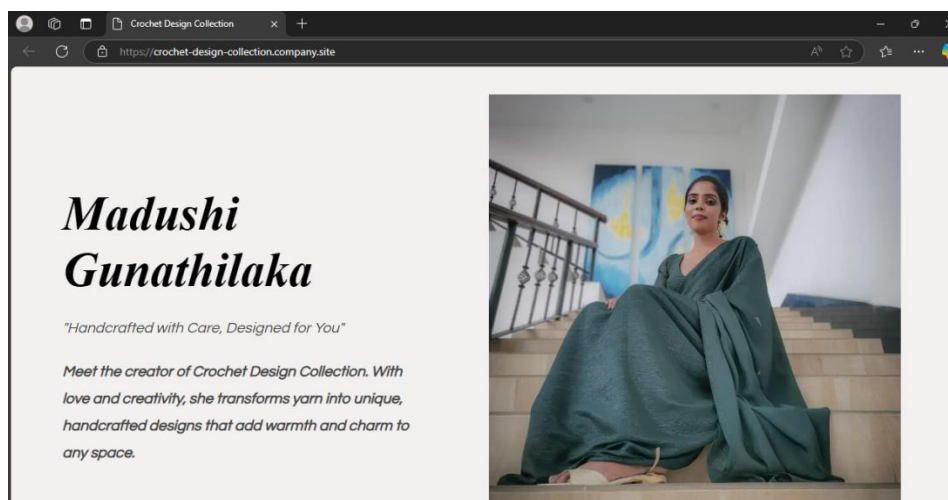


Figure 11- Business Owner

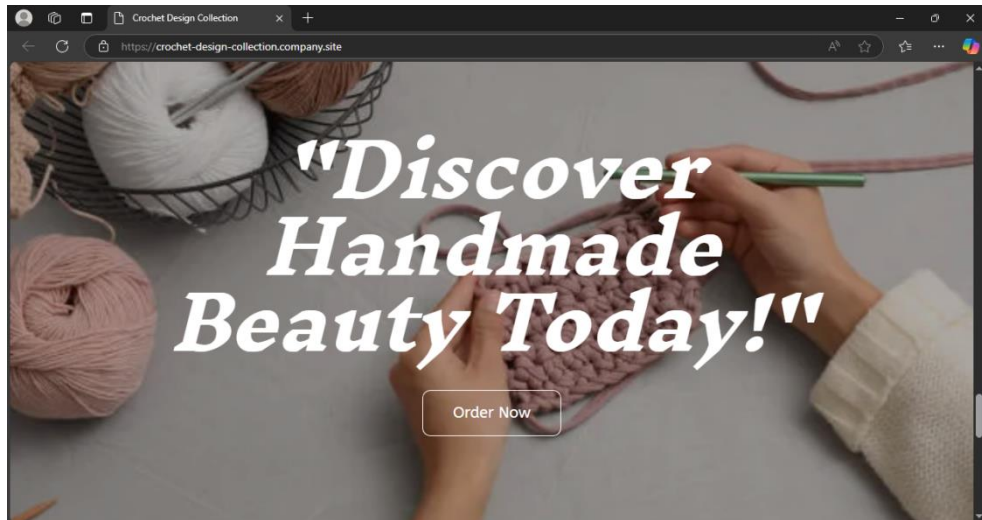


Figure 12- Order Page

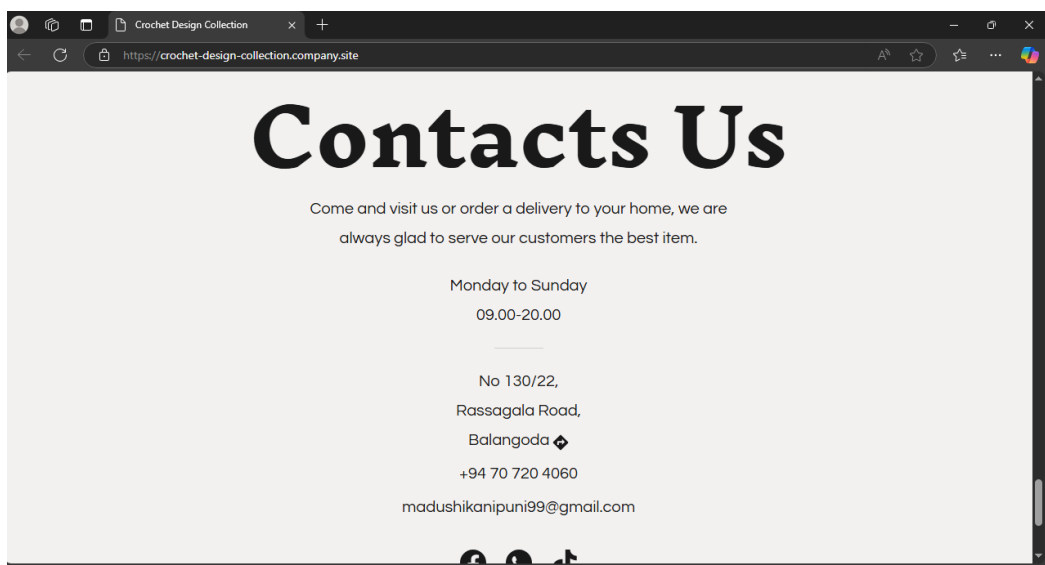


Figure 13- Contact Page

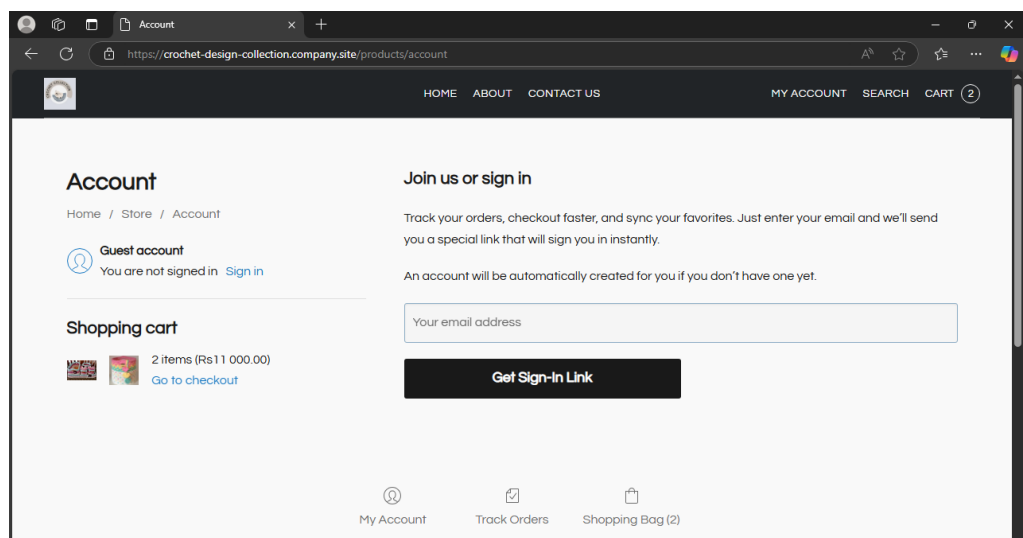


Figure 14- My Account Page

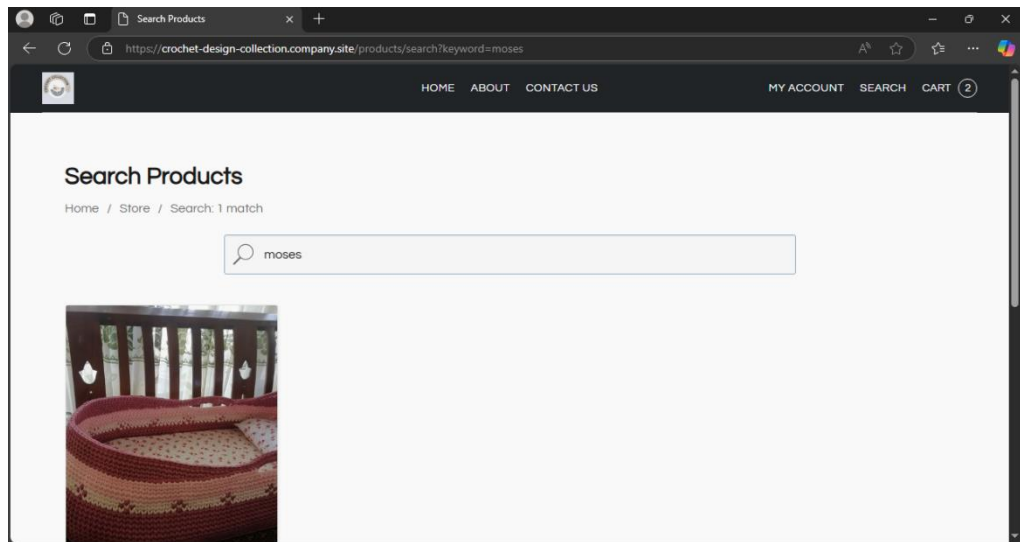


Figure 15- Search Page

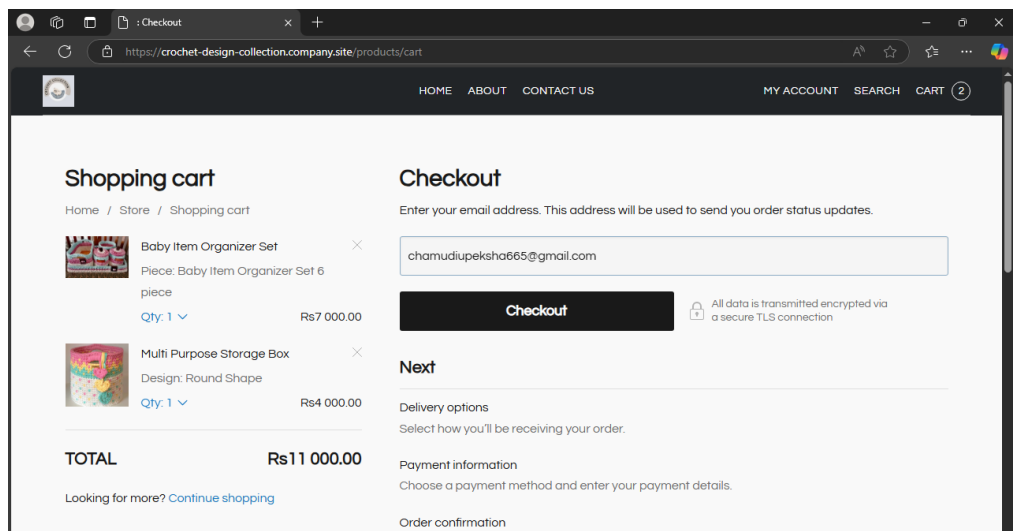


Figure 16- Cart Page

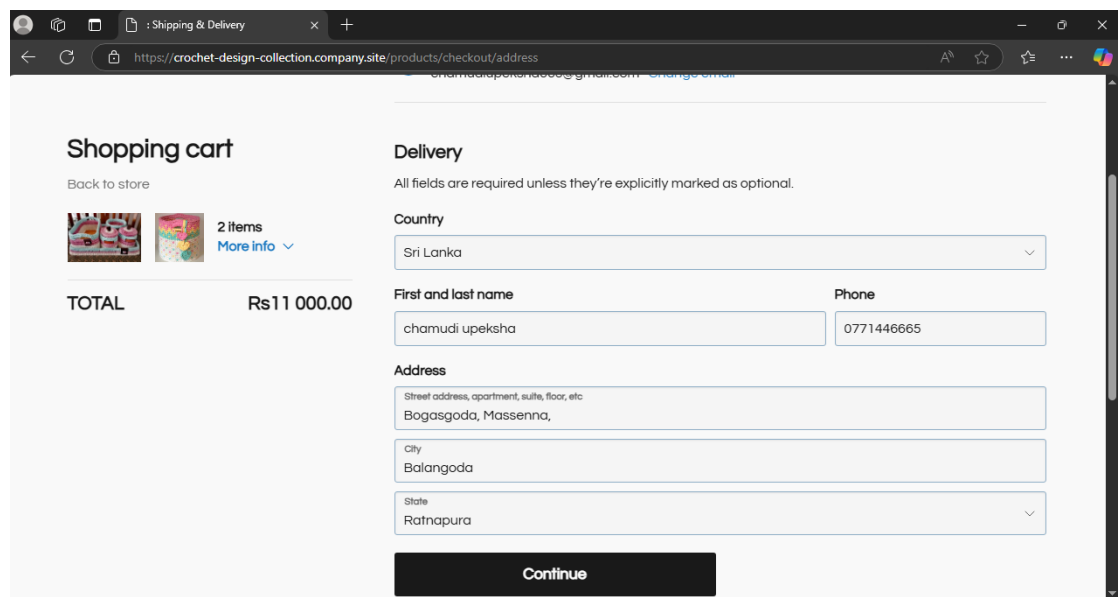
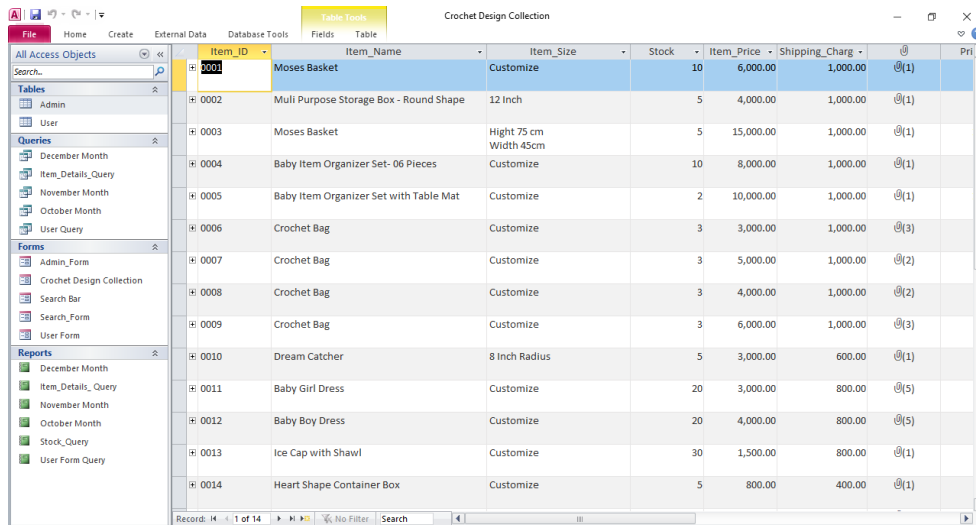


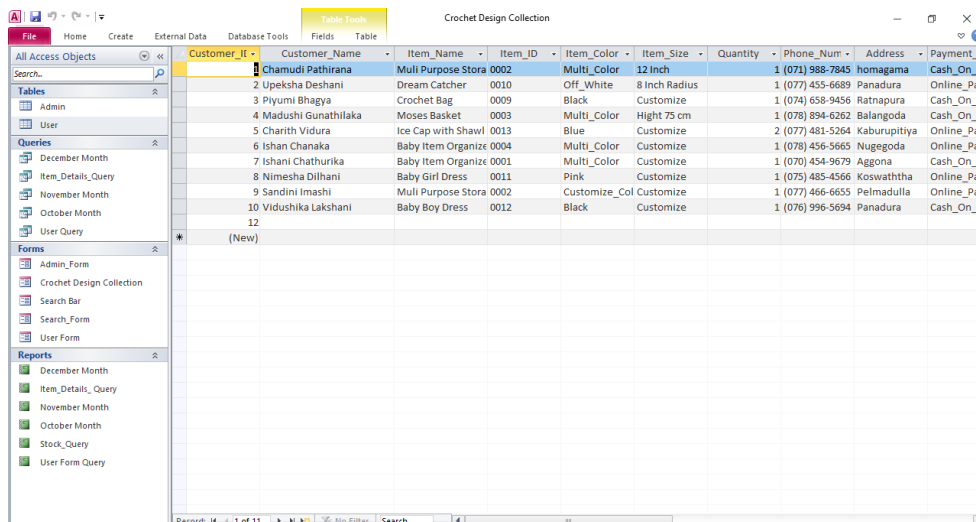
Figure 17- Back End

4.2.2 Data Base



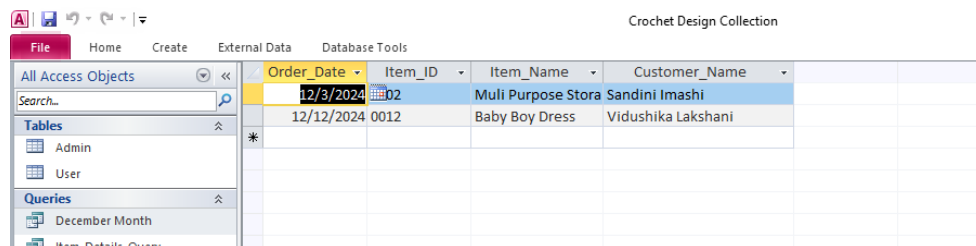
Item_ID	Item_Name	Item_Size	Stock	Item_Price	Shipping_Charg	Pri
0001	Moses Basket	Customize	10	6,000.00	1,000.00	@(1)
0002	Muli Purpose Storage Box - Round Shape	12 Inch	5	4,000.00	1,000.00	@(1)
0003	Moses Basket	Hight 75 cm Width 45cm	5	15,000.00	1,000.00	@(1)
0004	Baby Item Organizer Set- 06 Pieces	Customize	10	8,000.00	1,000.00	@(1)
0005	Baby Item Organizer Set with Table Mat	Customize	2	10,000.00	1,000.00	@(1)
0006	Crochet Bag	Customize	3	3,000.00	1,000.00	@(3)
0007	Crochet Bag	Customize	3	5,000.00	1,000.00	@(2)
0008	Crochet Bag	Customize	3	4,000.00	1,000.00	@(2)
0009	Crochet Bag	Customize	3	6,000.00	1,000.00	@(3)
0010	Dream Catcher	8 Inch Radius	5	3,000.00	600.00	@(1)
0011	Baby Girl Dress	Customize	20	3,000.00	800.00	@(5)
0012	Baby Boy Dress	Customize	20	4,000.00	800.00	@(5)
0013	Ice Cap with Shawl	Customize	30	1,500.00	800.00	@(1)
0014	Heart Shape Container Box	Customize	5	800.00	400.00	@(1)

Figure 18- Admin Table



Customer_ID	Customer_Name	Item_Name	Item_ID	Item_Color	Item_Size	Quantity	Phone_Num	Address	Payment_M
1	Chamudi Pathirana	Muli Purpose Stora	0002	Multi_Color	12 Inch	1	(071) 988-7845	homagama	Cash_On_Di
2	Upeksha Deshani	Dream Catcher	0010	Off_White	8 Inch Radius	1	(077) 455-6689	Panadura	Online_Pay
3	Piyumi Bhagya	Crochet Bag	0009	Black	Customize	1	(074) 658-9456	Ratnapura	Cash_On_Di
4	Madushi Gunathilaka	Moses Basket	0003	Multi_Color	Hight 75 cm	1	(078) 894-6262	Balangoda	Cash_On_Di
5	Charith Vidura	Ice Cap with Shawl	0013	Blue	Customize	2	(077) 481-5264	Kaburupitiya	Online_Pay
6	Ishan Chansaka	Baby Item Organiz	0004	Multi_Color	Customize	1	(078) 456-5665	Nugegoda	Online_Pay
7	Ishani Chathurika	Baby Item Organiz	0001	Multi_Color	Customize	1	(078) 454-9679	Aegona	Cash_On_Di
8	Nimesha Dillhani	Baby Girl Dress	0011	Pink	Customize	1	(075) 485-4566	Koswaththa	Online_Pay
9	Sandini Imashi	Muli Purpose Stora	0002	Customize_Col	Customize	1	(077) 466-6655	Pelmadulla	Online_Pay
10	Vidushika Lakshani	Baby Boy Dress	0012	Black	Customize	1	(076) 596-5694	Panadura	Cash_On_Di
12	(New)								

Figure 19- User Table



Order_Date	Item_ID	Item_Name	Customer_Name
12/3/2024	002	Muli Purpose Stora	Sandini Imashi
12/12/2024	0012	Baby Boy Dress	Vidushika Lakshani

Figure 20- December Month Query

Crochet Design Collection

Order_Date	Item_ID	Item_Name	Quantity	Customer_Name
10/1/2024	002	Muli Purpose Storage Box - Round	1	Chamudi Pathirana
10/16/2024	0010	Dream Catcher	1	Upeksha Deshani

Figure 21- November Month Query

Crochet Design Collection

Customer_Name	Quantity	Phone_Num	Address	Order_Date	Item_ID	Item_Name	Item_Size
Shani Chathurika	1	(070) 454-9679	Aggona	11/27/2024	0001	Moses Basket	Customize
Chamudi Pathirana	1	(071) 988-7845	homagama	10/1/2024	0002	Muli Purpose Storage Box - Round	12 Inch
Sandini Imashi	1	(077) 466-6655	Pelmadulla	12/3/2024	0002	Muli Purpose Storage Box - Round	12 Inch
Madushi Gunathilaka	1	(078) 894-6262	Balangoda	11/5/2024	0003	Moses Basket	Hight 75 cm
Ishan Chanaka	1	(078) 456-5665	Nugegoda	11/22/2024	0004	Baby Item Organizer Set- 06 Pieces	Customize
Piyumi Bhagya	1	(074) 658-9456	Ratnapura	9/27/2024	0009	Crochet Bag	Customize
Upeksha Deshani	1	(077) 455-6689	Panadura	10/16/2024	0010	Dream Catcher	8 Inch Radius
Nimesha Dilhani	1	(075) 485-4566	Koswaththa	11/30/2024	0011	Baby Girl Dress	Customize
Vidushika Lakshani	1	(076) 996-5694	Panadura	12/12/2024	0012	Baby Boy Dress	Customize
Charith Vidura	2	(077) 481-5264	Kaburupitiya	11/20/2024	0013	Ice Cap with Shawl	Customize

Figure 22- October Month Query

Crochet Design Collection

Item_ID	Item_Name	Item_Size	Price
0001	Moses Basket	Customize	7000.00
0002	Muli Purpose Storage Box - Round	12 Inch	5000.00
0003	Moses Basket	Hight 75 cm Width 45cm	16000.00
0004	Baby Item Organizer Set- 06 Pieces	Customize	9000.00
0005	Baby Item Organizer Set with Table Mat	Customize	11000.00
0006	Crochet Bag	Customize	4000.00
0007	Crochet Bag	Customize	6000.00
0008	Crochet Bag	Customize	5000.00
0009	Crochet Bag	Customize	7000.00
0010	Dream Catcher	8 Inch Radius	3600.00
0011	Baby Girl Dress	Customize	3800.00
0012	Baby Boy Dress	Customize	4800.00
0013	Ice Cap with Shawl	Customize	2300.00
0014	Heart Shape Container Box	Customize	1200.00

Figure 23- Item Details Query

Order_Date	Item_ID	Item_Name	Customer_Name
11/27/2024	0001	Moses Basket	Ishani Chaturika
11/5/2024	0003	Moses Basket	Madushi Gunathilaka
11/22/2024	0004	Baby Item Organizer Set- 06	Ishan Chanaka
11/30/2024	0011	Baby Girl Dress	Nimesha Dilhani
11/20/2024	0013	Ice Cap with Shawl	Charith Vidura

Figure 24- User Query

Admin Form

Item_ID: 0001

Item_Name: Moses Basket

Item_Size: Customize

Stock: 10

Item_Price: 6,000.00

Shipping_Charge: 1,000.00

Item Image:

Buttons: Add Record, Undo, Clear, Back, Navigation arrows.

Figure 25- Admin Form



Figure 26- Main Form

A Search Bar

Item_ID	Item_Name	Color	Item_Size	Customer_Name	Address	Contact_Nu	Customer_ID	Shipping_M

Figure 27- Search Bar

A Search_Form

SEARCH_BOX

1	Chamudi Pathi	Muli Purpose S	0002	Multi_Color	12 Inch	Cash_On_Delivery
2	Upeksha Desha	Dream Catcher	0010	Off_White	8 Inch Radius	Online_Payment
3	Piyumi Bhagya	Crochet Bag	0009	Black	Customize	Cash_On_Delivery
4	Madushi Gunat	Moses Basket	0003	Multi_Color	Hight 75 cm	Cash_On_Delivery
5	Charith Vidura	Ice Cap with SH	0013	Blue	Customize	Online_Payment
6	Ishan Chanaka	Baby Item Orga	0004	Multi_Color	Customize	Online_Payment
7	Ishani Chathuri	Baby Item Orga	0001	Multi_Color	Customize	Cash_On_Delivery
8	Nimesha Dilha	Baby Girl Dress	0011	Pink	Customize	Online_Payment
9	Sandini Imashi	Muli Purpose S	0002	Customize_Col	Customize	Online_Payment
10	Vidushika Laks	Baby Boy Dress	0012	Black	Customize	Cash On Delivery

Figure 28- Search Form

A User Form

User Form

Customer_Name

Ishani Chathurika

Item_Name

Moses Basket

Phone_Number

(070) 454-9679

Item_ID

0001

Address

Aggona

Item_Size

Customize

Quantity

1

Item_Color

Multi_Color

Payment_Method

Cash_On_Delivery

Price

\$7,000.00

Shipping_Method

Qurier

Order_Date

11/27/2024

Save

Delete

Undo

◀

▶

Figure 29- User Form

All Access Objects

Search...

Tables

Admin

User

Queries

December Month

Item_Details_Query

November Month

October Month

User Query

December Month

Thursday, January 30, 2025

1:32:13 AM

Order_Date	Item_ID	Item_Name	Customer_Name
12/3/2024	0002	Muli Purpose Storage Box - Round Shape	Sandini Imashi
12/12/2024	0012	Baby Boy Dress	Vidushika Lakshani

7

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Figure 30- December Month Report

User Query				Thursday, January 30, 2025	
				1:33:06 AM	
Item_ID	Item_Name	Item_Size	Price		
0001	Moses Basket	Customize	7000.00		
0002	Muli Purpose Storage Box - Round Shape	12 Inch	5000.00		
0003	Moses Basket	Hight 75 cm Width 45cm	16000.00		
0004	Baby Item Organizer Set- 06 Pieces	Customize	9000.00		
0005	Baby Item Organizer Set with Table Mat	Customize	11000.00		
0006	Crochet Bag	Customize	4000.00		
0007	Crochet Bag	Customize	6000.00		
0008	Crochet Bag	Customize	5000.00		
0009	Crochet Bag	Customize	7000.00		
0010	Dream Catcher	8 Inch Radius	3600.00		
0011	Baby Girl Dress	Customize	3800.00		
0012	Baby Boy Dress	Customize	4800.00		
0013	Ice Cap with Shawl	Customize	2300.00		
0014	Heart Shape Container Box	Customize	1200.00		
14				Page 1 of 1	

Figure 31- Item Details Report

November Month							Thursday, January 30, 2025	
							1:33:30 AM	
Customer_Name	Quantity	Phone_Number	Address	Order_Date	Item_ID	Item_Name		
Ishani Chathurika	1	(070) 454-9679	Aggona	11/27/2024	0001	Moses Basket		
Chamudi Pathirana	1	(071) 988-7845	homagama	10/1/2024	0002	Muli Purpose Storage Box - Round Shape		
Sandini Imashi	1	(077) 466-6655	Pelmadulla	12/3/2024	0002	Muli Purpose Storage Box - Round Shape		
Madushi Gunathilaka	1	(078) 894-6262	Balangoda	11/5/2024	0003	Moses Basket		
Ishan Chanaka	1	(078) 456-5665	Nugegoda	11/22/2024	0004	Baby Item Organizer Set- 06 Pieces		
Piyumi Bhagya	1	(074) 658-9456	Ratnapura	9/27/2024	0009	Crochet Bag		
Upeksha Deshani	1	(077) 455-6689	Panadura	10/16/2024	0010	Dream Catcher		
Nimasha Dilhani	1	(076) 495-4566	Korumbukke	11/20/2024	0011	Baby Girl Dress		

Figure 32- November Month Report

October Month		Thursday, January 30, 2025 1:33:50 AM		
Order_Date	Item_ID	Item_Name	Quantity	Customer_Name
10/1/2024	0002	Muli Purpose Storage Box - Round Shape	1	Chamudi Pathirana
10/16/2024	0010	Dream Catcher	1	Upeksha Deshani

Page 1 of 1

Figure 33- October Month Report

User Form Query		Thursday, January 30, 2025 1:34:10 AM				
Customer_Name	Quantity	Phone_Number	Address	Order_Date	Item_ID	Item_Name
Ishani Chathurika	1	(070) 454-9679	Aggona	11/27/2024	0001	Moses Basket
Chamudi Pathirana	1	(071) 988-7845	homagama	10/1/2024	0002	Muli Purpose Storag Box - Round Shape
Sandini Imashi	1	(077) 466-6655	Pelmadulla	12/3/2024	0002	Muli Purpose Storag Box - Round Shape
Madushi Gunathilaka	1	(078) 894-6262	Balangoda	11/5/2024	0003	Moses Basket
Ishan Chanaka	1	(078) 456-5665	Nugegoda	11/22/2024	0004	Baby Item Organizer Set- 06 Pieces
Piyumi Bhagya	1	(074) 658-9456	Ratnapura	9/27/2024	0009	Crochet Bag
Upeksha Deshani	1	(077) 455-6689	Panadura	10/16/2024	0010	Dream Catcher
Nimesha Dilhani	1	(075) 485-4566	Koswaththa	11/30/2024	0011	Baby Girl Dress
Vidushika Lakshani	1	(076) 996-5694	Panadura	12/12/2024	0012	Baby Boy Dress
Charith Vidura	2	(077) 481-5264	Kaburupitiya	11/20/2024	0013	Ice Cap with Shawl

Figure 34- User Query Report

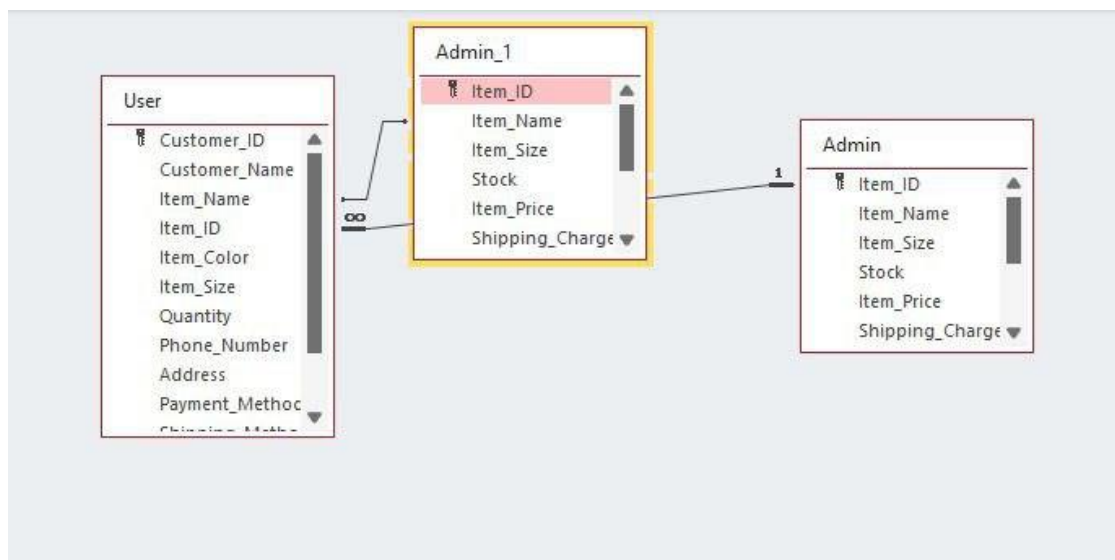


Figure 35- Relationship

4.2.3 Data Base System

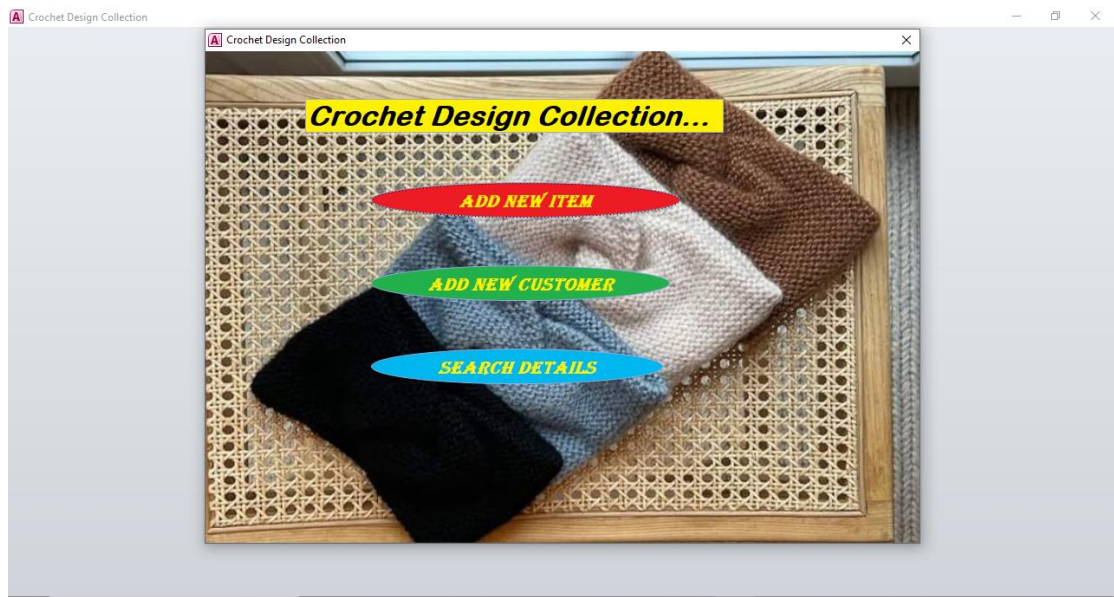


Figure 36- Data Base System

Chapter 5: Results and Testing

5.1 Final Outcome

The Crochet Design Collection now has a professional and fully functional online store thanks to the project's successful completion. With scalability in mind, the store was created using Ecwid as the main e-commerce administration platform. Ecwid's integration made it possible to handle customer interactions, order processing, and the product catalog with ease.

All product data, customer information, and transaction records are stored and managed in large part by the back-end database, which is constructed using Microsoft Access. It guarantees effective inventory management, retrieval, and updating while providing the flexibility needed to expand as the company does. Order fulfillment runs smoothly thanks to the database system, which also makes sure that clients are given accurate information about product availability and that stock levels are updated in real time.

With an intuitive layout and simple navigation, the website was created to offer a seamless and enjoyable user experience. Because the design was responsively tuned, it works well on a variety of platforms, such as desktop computers, tablets, and smartphones. Customers may buy products with ease thanks to the simplified checkout process, and secure and dependable transactions are guaranteed by payment integration.

Important elements including a product search tool, customer login sections, and an interactive interface that lets users peruse a range of crochet designs are also supported by the website. Ecwid's interface with cutting-edge database systems guarantees that the website can readily handle new additions as the product catalog grows without compromising usability or speed.

In conclusion, Ecwid's e-commerce technologies are effectively integrated with a strong MS Access database system to create a polished, scalable, and user-friendly online store that offers a smooth shopping experience for customers and a strong basis for future expansion.

5.2 Testing and Feedback

5.2.1 Testing

"Compared to other technologies, software has more frequent, widespread, and problematic errors."

The process of software testing helps determine whether developed computer software is accurate, comprehensive, secure, and of high quality. Accordingly, testing can never fully prove that an arbitrary computer program is correct.

Although software testing can be approached in a variety of ways, testing complicated products effectively basically involves research.

Examine and verify how the system is represented in the design, requirements document, diagrams, and program source code. They can be used during the entire process.

We employ three different kinds of testing. They are as follows:

5.2.1.1 Unit Testing

Individual component testing is typically the initial step in the testing process for software applications. Testing distinct system components separately is known as unit testing.

5.2.1.2 Integration Testing

The components are only merged and tested at every level of integration after each component has been tested separately. Integration testing is the term for that.

5.2.1.3 System Testing

System testing is the process of testing the fully integrated system in the end.

5.2.2 Feedback

Feedback from technical teams and users during the testing stages pointed out a number of areas that needed work. The most important comments were on improving mobile user navigation, speeding up the website's load time, and streamlining the checkout process for easier transactions. Based on usability testing, user experience enhancements were implemented to make sure the interface was user-friendly and compatible with all devices.

Performance testing showed that to increase page loading performance, little database query and picture compression optimizations were required. To guarantee smooth data synchronization, especially for real-time inventory management and order processing, integration problems between the Ecwid platform and MS Access were fixed.

With encryption techniques and secure payment channels guaranteeing the safety of consumer data, security audits validated the effectiveness of data protection measures. As the website grows, post-launch input will continue to direct additional improvements, with a focus on preserving system effectiveness and boosting user engagement.

Chapter 6: Conclusion

6.1 Summary

This project has effectively created a dynamic, user-focused online store that combines the strong capabilities of a sophisticated MS Access database with the e-commerce platform power of Ecwid. The end product is a professional, scalable, and incredibly effective web shop for maintaining and exhibiting crochet creations.

A seamless, enjoyable shopping experience is guaranteed by the user-friendly UI/UX design, which enables consumers to browse the product catalog with ease, make safe purchases, and take advantage of a responsive interface on all platforms. Real-time inventory updates, customer tracking, and efficient order administration are all powered by the MS Access database behind the scenes, guaranteeing that the company may grow without sacrificing quality.

By including these components, the platform improves operational efficiency and raises the legitimacy of the Crochet Design Collection while providing customers with a sophisticated buying experience and a smooth backend for store management.

6.2 Future Improvements

A number of cutting-edge features and optimizations can be added to the Crochet Design Collection website to further improve user experience, expedite corporate processes, and broaden the market. Key upcoming implementations are listed below:

1. AI-Powered Suggestions & Personalization

Use machine learning algorithms to examine consumer behavior and offer tailored product recommendations based on browsing habits, past purchases, and personal preferences.

AI-powered chatbots can provide immediate customer service, respond to questions, and help with the purchasing process.

2. Order and Inventory Management Automation

Real-time stock changes to guarantee smooth order fulfillment and avoid overselling. Automating restocking based on sales data and integrating vendor and supplier management systems.

Utilize predictive analytics to forecast demand and maximize stock levels.

3. Advanced Features for Payment and Security

Payment processing using blockchain technology to guarantee safe and open transactions.

Support for bitcoin payments allows for low transaction costs when making purchases abroad.

For safe login and payment, use biometric authentication (facial recognition or fingerprint).

4. Mobile optimization and progressive web applications (PWAs)

Create a PWA version so that consumers may install the website as an app without having to download it from the app store.

Improved mobile-first design for easy tablet and smartphone shopping.

5. Integration of Augmented Reality (AR)

Before making a purchase, clients can view a virtual product simulation that shows how the crocheted products will appear in real life.

AR-guided custom order lessons that let users interactively request unique crochet patterns.

6. Membership & Subscription Model

Establish a loyalty program that provides members with free shipping, early access to new designs, and exclusive pricing.

Customers receive subscription boxes or monthly crochet kits based on their preferences.

7. International Development & Multilingual Assistance

Provide multilingual and multicurrency support for clients from across the world.

Automate customs processing and anticipated delivery times by integrating international shipping options.

8. Influencer & Social Commerce Partnerships

Turn on social media shopping integration so that direct purchases from Facebook, Pinterest, and Instagram are possible.

Collaborate with craft bloggers and influencers to market unique crochet collections.

Chapter 7: Reference

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